

Computer Science graduate with hands-on experience building AI-powered full-stack applications, scalable backend systems, and production-ready AI services. Skilled in Python, FastAPI, React, Node.js, Docker, AWS, DigitalOcean, PostgreSQL, MongoDB, and Qdrant. Experienced designing and deploying distributed, cloud-native applications across multi-cloud environments using containerization and secure networking.

Work Experience

Backend Developer Intern Jul 2025 - Sep 2025
Common Criteria Pakistan | Islamabad, Pakistan (On-Site)

- Designed and deployed 5+ Laravel backend modules, optimizing complex SQL querying and logic to improve overall system response time by ~20%.
- Co-developed an internal HRMS and Document Management System (DMS), engineering a recursive directory upload feature that enables direct folder ingestion (bypassing standard RAR compression), reducing manual file handling by 40%.
- Collaborated in a shared codebase using Git, conducting peer code reviews and enforcing version control standards to decrease post-deployment bugs by 15%.

Projects

VisionIndex Video Analytics Platform (Final Year Project) Nov 2025 - Apr 2026

Developed a distributed full-stack video analytics platform using React, Node.js, Express, PostgreSQL, MongoDB, and Qdrant. Implemented REST APIs, role-based authentication, video upload workflows, and search APIs to support efficient retrieval of indexed surveillance footage. Designed and integrated a Python-based AI processing service using Temporal workflows for asynchronous request handling, enabling person and object detection, tracking, embedding generation, vector indexing in Qdrant, and multimodal semantic search. Containerized application services using Docker and deployed a multi-cloud architecture, hosting the AI processing pipeline on an AWS EC2 GPU instance (g4dn) while deploying the frontend, backend, PostgreSQL, MongoDB, and Qdrant on a DigitalOcean Droplet. Established secure inter-cloud communication between AWS and DigitalOcean using Tailscale and integrated Amazon S3 for scalable video storage. Awarded PKR 30,000 in ORIC funding in recognition of the project's innovation and commercialization potential.

AUIS Nexus Management System Mar 2025 - May 2025

Developed a full-stack MERN platform used by a university student society with 20+ members, featuring secure authentication, Role-Based Access Control (RBAC), event management, and blog functionalities. Built scalable RESTful APIs, an Admin Dashboard for user and content management, and optimized MongoDB indexing to streamline community operations and backend workflows.

SpotTube Automated Content Synchronizer May 2024 - Jun 2024

Engineered a Python ETL pipeline and automation script to sync playlists by cross-referencing Spotify and YouTube APIs. Utilized multiprocessing to optimize concurrent data processing, reducing batch download times by ~40% compared to sequential fetching.

Core Skills

Languages: Python, JavaScript, PHP, C++
Backend Frameworks: FastAPI, Express.js, Flask, Laravel
Frontend: React, HTML5, CSS3, Responsive Design
Databases: PostgreSQL, MySQL, MongoDB, Qdrant
Cloud & DevOps: AWS EC2, AWS S3, DigitalOcean Droplets, Docker, Tailscale, Linux, Git, Postman, n8n
Software Engineering: REST APIs, OOP, Data Structures, Algorithms, SDLC, Agile/Scrum, Clean Code, Debugging
AI & Machine Learning: Machine Learning, Deep Learning, Computer Vision, CLIP, Vector Search, Semantic Search, ETL, Pandas, NumPy

Education

Air University 2022 - 2026
Bachelor Computer Science
GPA: 3.15 / 4.00

Certificates

Supervised Machine Learning (DeepLearning.AI via Coursera) Nov 2025
Generative AI Application Developer (NCEAC) Mar 2026